

Octo · arXiv 2405.12213 · RSS 2024

Stakeholder

18+3 · seat 3

Open FT ancestor, not 2026 SOTA
Keep as eval baseline · ship $\pi_{0.5}$ /GR00T

Oct 14 paper club · VLA Architectures · Oct 16 BEHAVIOR nearby

Spine: open OXE GRP + diffusion chunks · underspecified horizons · NOT 2026 winner weights

Octo Model Team · Berkeley / Stanford / CMU / DeepMind · HF rail-berkeley/octo-*

Club line

**ACT || DP → Octo (open OXE GRP)
→ π_0 / OpenVLA → OpenPI / BEHAVIOR**

π_0 (CLEAR) beats Octo-93M on hard dexterous tasks with VLM+FM.
Octo = infrastructure history + eval baseline — not 2026 winner weights.

Prefer $\pi_{0.5}$ / N1.7 for BEHAVIOR · keep Octo for open FT ancestor + mixture lessons

Stakeholder value · open FT ancestor, not product brain

Open usable GRP

27M / 93M HF ckpts
JAX FT scripts + OXE loaders
2024 default “try a pretrained policy”

Cost model that sells

~100 demos + ~5 h A5000
→ 72% avg vs 20% scratch
Consumer-GPU honest

NOT 2026 SOTA

No Internet VLM backbone
93M vs π_0 ~3B
 π_0 CLEAR-beats hard dexterity

Strategic ask

Keep Octo in eval harness
Ship $\pi_{0.5}$ / GR00T for BEHAVIOR'26
Oct 16 deadline nearby

Leadership one-liner

Octo proved open cross-embodiment pretrain + cheap FT works;
 π_0 proved you still need VLM scale + better generative action modeling for household dexterity.

Numbers leadership remembers

800k

demos · 25 OXE sets

27M / 93M

open checkpoints

+29 pp

ZS vs RT-1-X

72%

FT avg @ ~100 demos

~5 h

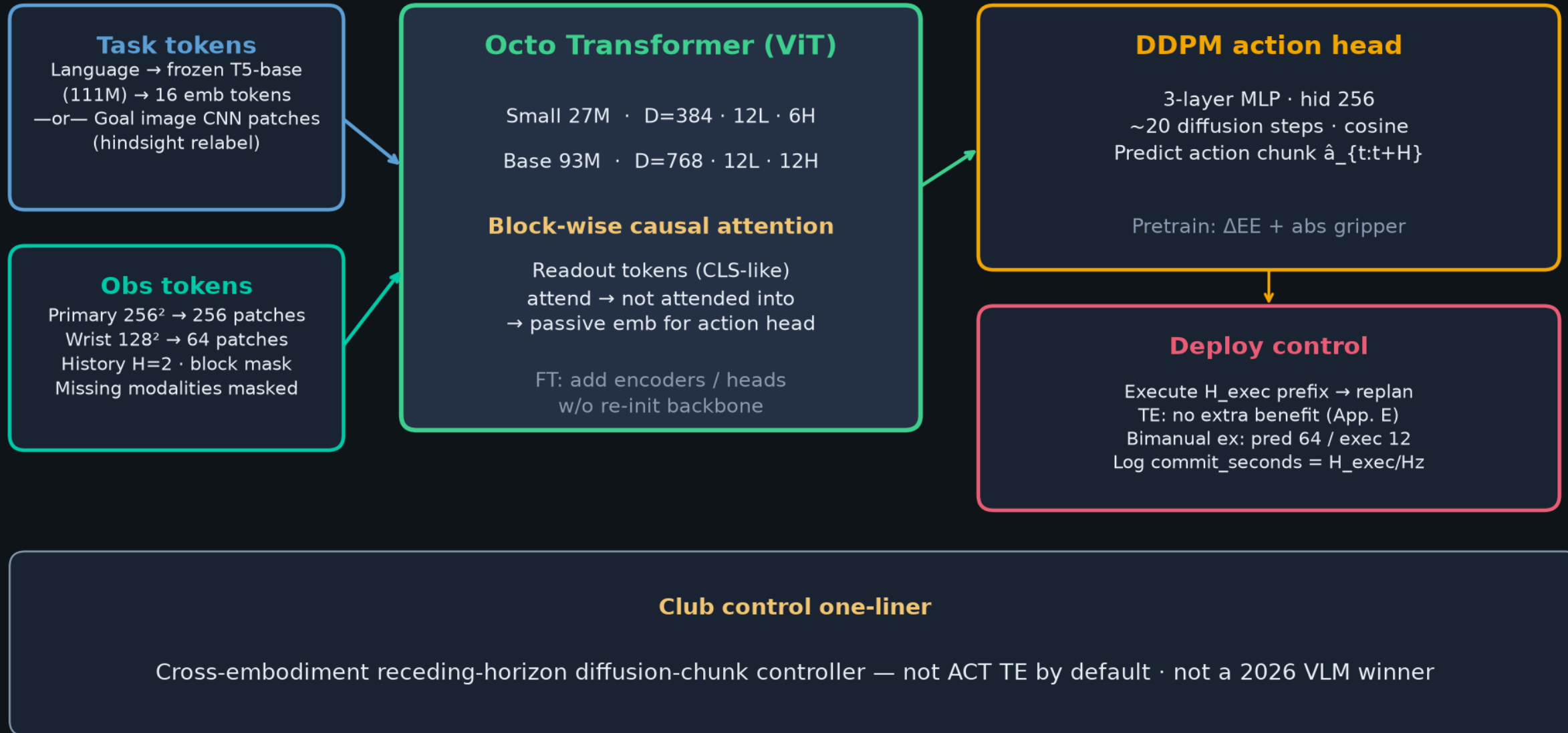
A5000 FT walltime

9 robots

4 institutions

Octo stack · ViT + frozen T5 + DDPM chunk head

Single backbone forward · light head denoises · receding-horizon execute (TE not helpful)



Octo: An Open-Source Generalist Robot Policy

Octo Model Team

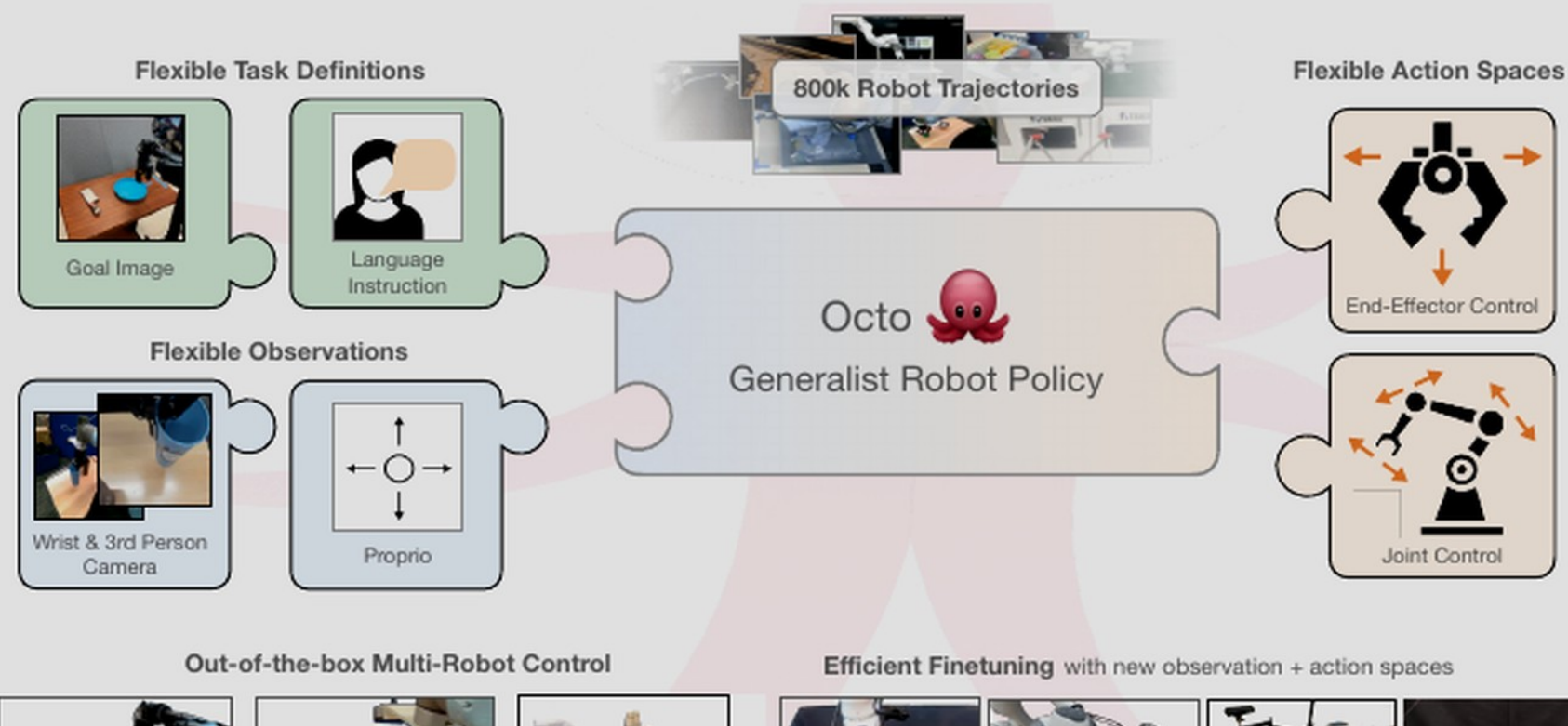
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<https://octo-models.github.io>



FT recipe card · operationalizable (rare honesty)

Data

~100 target demos

Steps

50k · cosine + warmup

Update

Full model (beat freeze)

Compute

~5 h · A5000 24 GB

Adapters

New cam / F/T / joint / 14-DoF

Result

Table I avg 72% vs 20%/15%

Domains: Insertion* (F/T) · Coffee · Baking · Pick-Up† · Coke · Bimanual† (*new obs †new action)

scratch as well as state-of-the-art pretrained visual representations. Each domain uses ~ 100 target demonstrations and the same finetuning hyperparameters. In each domain, success rates are averaged over 20 trials. *: New observation input (force-torque proprioception). †: New action space (joint position control).

placing, wiping a table with a cloth, and opening and closing drawers. For each robot, we selected two language tasks from the corresponding OXE dataset and performed 10 trials per task with varying initial conditions (details in Appendix F). The chosen tasks are “in-distribution” from the pre-training data, but the evaluation requires methods to generalize to new object positions, lighting conditions, backgrounds, and distractor objects. While all methods acted reasonably across tasks in the pretraining environments, we found that on average Octo had a 29% higher success rate than RT-1-X (35M parameters). For the WidowX and RT-1 Robot evaluations, we also compared to RT-2-X (55 billion parameters) [103] and found that Octo performed similarly.

Additionally, while RT-1-X and RT-2-X only support conditioning on language instructions, Octo also supports conditioning on goal images. We evaluated our model on the WidowX tasks using goal image conditioning and found that it achieved a 25% higher success rate than when evaluated with language conditioning. This is likely because goal images provide more information about how to achieve the task. In the BridgeV2 domain, we performed a fine-grained analysis of the zero-shot

		Aggregate Performance
DATA	Octo-Small (Ours)	83%
	RT-X dataset mix [67]	60%
	Single robot dataset (Bridge Data)	43%
POLICY	Discretized Action Prediction [67]	18%
	Continuous Action Prediction (MSE)	35%
ARCH	Resnet-50 + Transformer[67]	70%

TABLE II: **Model Ablations.** We achieve best performance when using the ViT architecture, diffusion action head, and wide training data mixture. All evaluations are performed on the WidowX setup. Success rates are averaged over 40 trials across two language-conditioned tasks and two goal-conditioned tasks.

C. Design Decisions for Generalist Robot Policy Training

We have demonstrated the effectiveness of Octo as a zero-shot multi-robot controller and as an initialization for policy finetuning. We next analyze the effects of different design

$x^k \sim \mathcal{N}(0, I)$ and apply k steps of denoising with a learned denoising network $\epsilon_\theta(x^k, e, k)$ that is conditioned on the output x^k of the previous denoising step, the step index k , and the output embedding e of the transformer action readout:

$$x^{k-1} = \alpha(x^k - \gamma\epsilon_\theta(x^k, e, k) + \mathcal{N}(0, \sigma^2 I)). \quad (1)$$

The hyperparameters α , γ , and σ correspond to the noise schedule: we use the standard cosine schedule from [66]. We train the diffusion head using the standard DDPM objective first proposed in [34], where we add Gaussian noise to the dataset actions and train the denoising network $\epsilon_\theta(x^k, e, k)$ to reconstruct the original action. For a detailed explanation of diffusion policy training, see Chi et al. [17]. We list all hyperparameters in Appendix D.

We use the same diffusion training objective during finetuning and update the full model, a recipe which outperformed those that freeze subsets of the pretrained parameters. In all finetuning experiments, we employ the same recipe: given a small target domain dataset with around 100 trajectories, we finetune for 50k steps using a cosine decay learning rate decay with linear warmup.

D. Training Details

We trained two variants of our model: Octo-Small with a transformer backbone that mirrors the size of a ViT-S, and Octo-Base with a transformer backbone that mirrors the size of a ViT-B [22].

We use the AdamW optimizer [51] with an inverse square root decay learning rate schedule [97], with weight decay of

- **Finetuning scripts** for Octo models, in JAX.
- **Model pretraining pipeline** for Octo pretraining on the Open X-Embodiment dataset, in JAX.
- **Standalone data loaders** for Open X-Embodiment data, compatible with JAX and PyTorch.

We provide a simple example for loading and running a pretrained Octo model in Appendix B.

IV. EXPERIMENTS

Our experiments provide an empirical analysis of Octo, evaluating its ability to serve as a general robotic foundation model across several axes:

- 1) Can Octo control multiple robot embodiments and solve language and goal tasks out of the box?
- 2) Do Octo weights serve as a good initialization for data-efficient finetuning to new tasks and robots, and does it improve over training from scratch and commonly used pretrained representations?
- 3) Which design decisions in Octo matter most for building generalist robot policies?

Evaluation setups: We evaluate Octo’s capabilities across a representative spectrum of 9 robot learning setups at 4 institutions (see Fig. 4). We test Octo’s ability to control different robots out-of-the-box (“zero-shot”) for language and goal image tasks using robot setups that match the pretraining data, where all robots are controlled with delta end-effector control actions and the observation spaces are RGB images. We also evaluate Octo for data-efficient finetuning to new environments and

Octo on 9 real robot setups across 4 institutions. Our evaluations capture diverse object long task horizons (e.g., “Stanford Coffee”) and precise manipulation (e.g., “Berkeley capabilities to control robots in environments from the pretraining data out-of-the-box and environments with small target domain datasets. We also test finetuning with new Berkeley Peg Insertion”), action spaces (joint position control in “Berkeley Pick-Up” and bodiments (e.g., “Berkeley Bimanual” and “Berkeley Coke”).

ise manipulation (e.g., each evaluation setup,

’s ability to control best openly available using the released pretrained on the Open to control multiple point of comparison. ities of Octo to **RT-** mage model finetuned produce robot actions. re trained on a more are to 800k episodes performance as a policy ng to two common main demonstrations isual representations. osed other pretraining [26], to our knowledge *policy* that has been new observation and representations such

Majumdar et al. [57]. A ViT-B visual encoder is initialized to the VC-1 weights [57], a state-of-the-art visual representation pretrained on 4,000 hours of ego-centric videos and ImageNet, and combined with an MLP action decoder. The full model is trained to predict expert actions using an MSE loss (“VC-1”).

A. Octo Controls Multiple Robots Out-of-the-Box

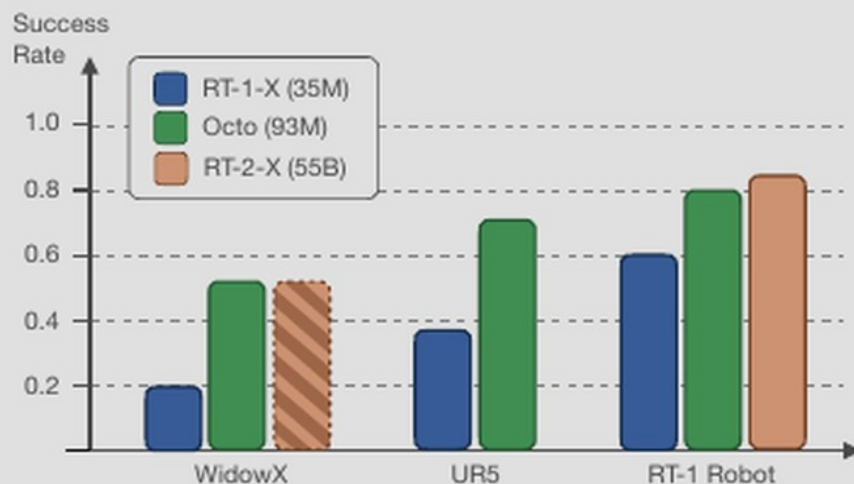


Fig. 5: **Zero-Shot Evaluation.** Out-of-the-box, Octo can control multiple robots in environments from the pretraining data.

Fig. 3 · Training mix — 25 OXE datasets

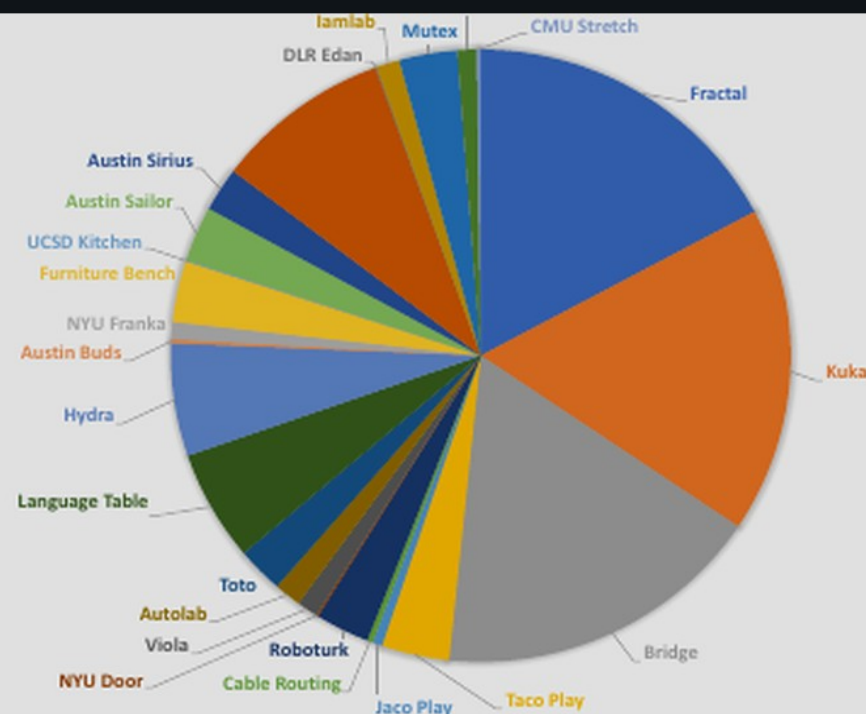
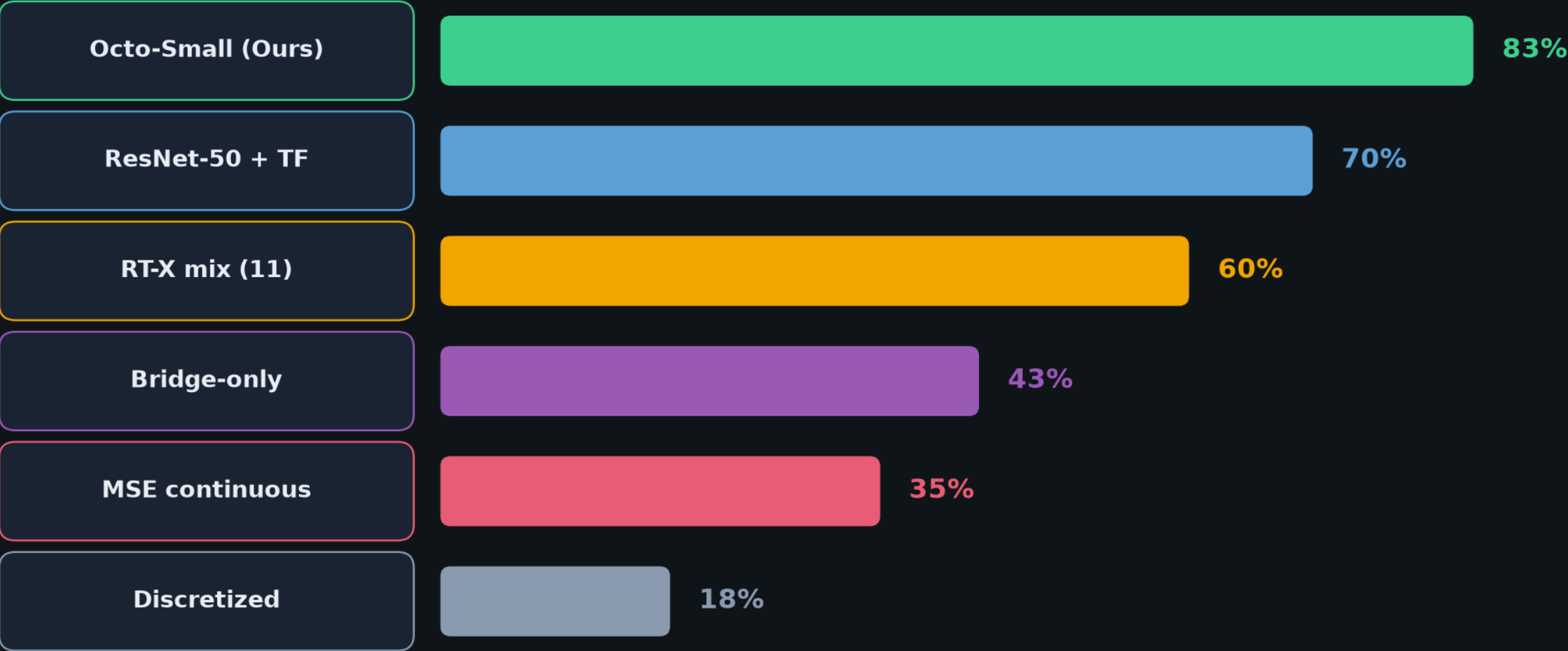


Fig. 3: **Training dataset composition.** We curate a subset of 25 datasets from the Open X-Embodiment dataset that have image observations, end-effector actions, and show diverse behaviors. The pie chart visualizes the fractions that each dataset contributes to every training batch on average. The dataset weights are determined by the number of samples in each dataset with small modifications to balance dataset size and diversity (see [Section III-B](#) for details).

action configurations during pretraining, being able to adapt Octo's inputs and outputs during finetuning makes it a versatile tool for the robotics community. Prior model designs that use

Ablation board · Table II WidowX Octo-Small



Diffusion >> MSE/discrete · mix width 43→60→83 · ViT-first helps at pretrain scale

Genealogy · CLEAR club line



Pivot: open ViT + diffusion-chunk GRP on 800k OXE with block-wise adapters

Octo = infrastructure history + eval baseline · NOT 2026 winner weights

Priors: OXE/RT-X · DP · ACT · ViT · T5 · DDPM · HER goals · RT-1/RoboCat · Bridge

Subsequent CLEAR: OpenVLA · π_0 (2410.24164) · OpenPI · $\pi_{0.5}$ /RTC · BEHAVIOR'25-'26

Underspecified horizons · motion FLAG

Main text: “several” future actions · only concrete: bimanual pred 64 / exec 12

System	Chunk object	Generative	Ensemble?	Typical commit
ACT	k×14 joints	CVAE	Yes TE	ensembled @ 50 Hz
Diffusion Policy	Tp actions	DDPM/DDIM	warm / Ta	Ta≈8 @ ~10 Hz
Octo (this)	action chunk	DDPM readout	No (paper)	receding prefix
π_0 / $\pi_{0.5}$	H actions	Flow matching	No	open-loop K @ 20-50
BEHAVIOR'25	H≈30-32	FM + corr.	soft inpaint	~30 Hz receding

Empiricist: always log (H_pred, H_exec, Hz, diff_steps) · commit_seconds = H_exec/Hz · TE not dead globally

Generalization honesty · Table VII WidowX



Cite 5% novel skill whenever someone says “generalist zero-shot”

Objects transfer; environments degrade; skills near floor — prevents foundation-model overclaim

Citation guidance · how Kanta should talk about Octo

USE Octo as...

- ✓ Canonical open OXE GRP + finetune-first API
- ✓ Evidence diffusion chunks $\gg$ MSE/bins at multi-embod. scale
- ✓ Case study: mixture curation $\approx$ architecture
- ✓ Ancestor bridge into π_0 / OpenPI / BEHAVIOR stacks
- ✓ Operationalizable FT recipe (~100 demos / 24 GB)

AVOID using Octo as...

- ✗ Proof that open 93M $\approx$ closed 55B VLA generally
- ✗ Evidence that TE is obsolete in all regimes
- ✗ Fully solved “any robot, any sensor” FM
- ✗ Drop-in 2026 BEHAVIOR baseline
- ✗ Prefer $\pi_{0.5}$ / N1.7 lineage for production

VERDICT

Strong RSS 2024 open systems paper between ACT/DP and π_0 /OpenVLA.
Praise openness + FT protocol; punish zero-shot overclaim and underspecified horizons.

Octo = infrastructure history + eval baseline · NOT 2026 winner weights · NOT BEHAVIOR drop-in

Stakeholder · close

Open FT ancestor — keep in eval harness **Ship $\pi_{0.5}$ / GROOT for BEHAVIOR**

Octo proved open cross-embod. pretrain + cheap FT
 π_0 proved you still need VLM scale + better generative actions

Q&A · keep the spine clear

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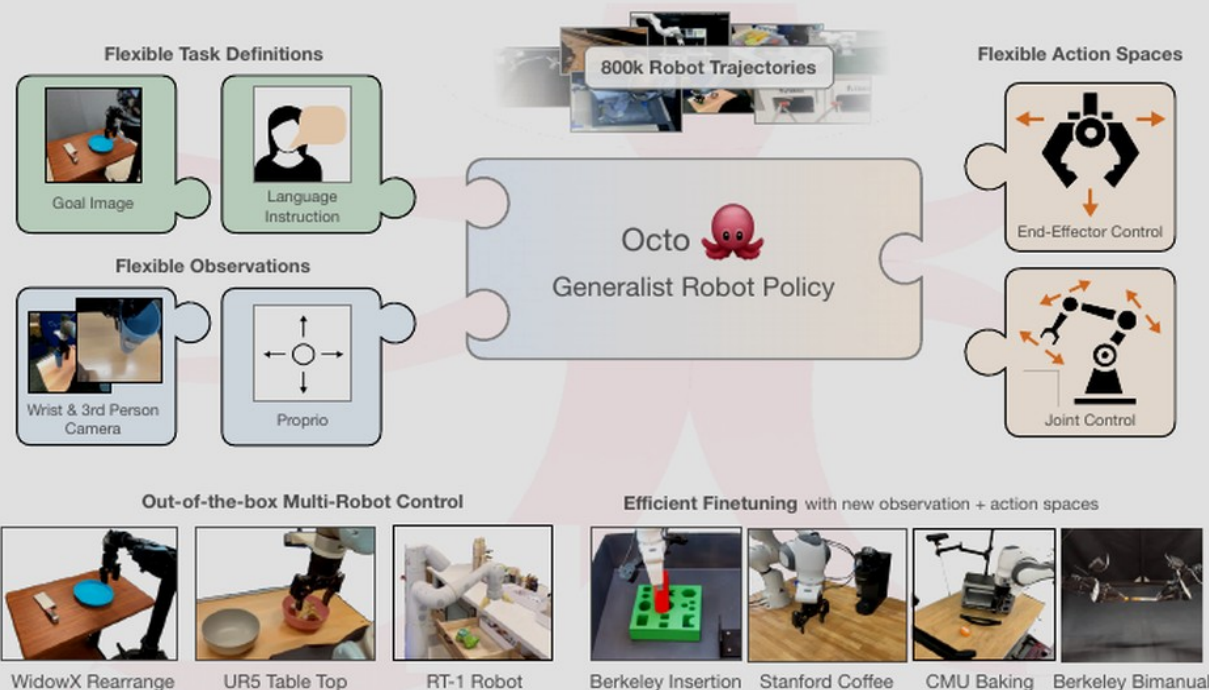


Fig. 1: We introduce Octo, an open-source, generalist policy for robotic manipulation. Octo is a transformer-based policy pretrained on 800k diverse robot episodes from the Open X-Embodiment dataset [67]. It supports flexible task and observation definitions and can be quickly finetuned to new observation and action spaces.

Abstract—Large policies pretrained on diverse robot datasets have the potential to transform robotic learning: instead of experiments across 9 robotic platforms, we demonstrate that Octo serves as a versatile policy initialization that can be effectively